MALLA REDDY UNIVERSITY

Maisammaguda, Kompally, Malkajgiri, Medchal District, Secunderabad 500100, Telangana, India
Email: mruh@mallareddyuniversity.ac.in, www.mallareddyuniversity.ac.in

IV Year B.Tech-I Semester Major Project Phase I Summary Sheet

Project Title:	Cyber Attack Prediction: From Traditional Machine Learning to Generative Artificial Intelligence				
Project Code:		**Batch Size:**	04	**Batch:**	2022-26
Domain / Area:	Cyber Security,Gen-AI	**SDG Mapping**	Industry , Innovation, Technology and Infrastructure		

Abstract:

This project, Cyber Attack Prediction: From Traditional Machine Learning to Generative Artificial Intelligence, explores the transition from conventional ML approaches to advanced Generative AI models in predicting and mitigating cyber attacks. The study compares the performance of classical algorithms such as Decision Trees, Random Forests, and Support Vector Machines with cutting-edge Generative AI frameworks, including Generative Adversarial Networks (GANs) and Transformer-based architectures.

Technical (S/w & H/w) Specifications

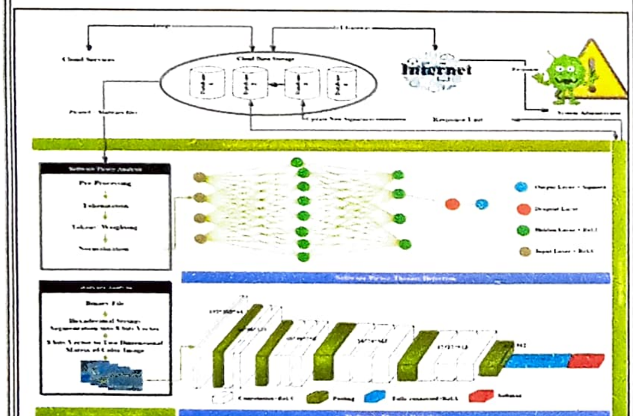
Software

- OS: Windows 10/11 or Ubuntu 20.04
- Languages: Python 3.8+
- Libraries: Scikit-learn, XGBoost, TensorFlow/PyTorch, Hugging Face

Module(s) Specifications

Data Acquisition Module
Preprocessing Module
Traditional ML Module
Generative AI Module

Architecture Diagram



Methodology

- Collect and preprocess cyber-attack datasets, handling missing values and normalization.
- Extract key features and build baseline ML models (SVM, RF, XGBoost).
- Apply Generative AI (Transformers/Autoencoders) for anomaly & zero-day attack detection.
- Evaluate models, visualize results, and deploy prediction system via web/API.

Existing System

1. Mostly relies on signature-based detection
2. Rule-based and static ML models struggle with dynamic threats.
3. Requires manual feature engineering and expert intervention.

Proposed System

1. Combines Traditional ML with Generative AI for robust predictions.
2. Automated feature learning using deep learning and transformers.
3. Detects novel/zero-day attacks via anomaly pattern

Guide Details

Dr.MD Adam Baba
Ph.D
Assistant Professor
Department of
Artificial Intelligence
& Machine Learning
Malla Reddy
University.

Batch Members Details

			
2211CS020361	2211CS020415	2211CS020527	2211CS020529